Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 3

GROUND STATE ENERGIES

E0 (Ha) : -230.75351694
E1 (Ha) : -230.57821527
Gap (Ha) : 0.17530166
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,5)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 5
Spin-orbitals   : 10
Hilbert dim   : 100

SPIN PROPERTIES

<S2>   : -0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.961507
<D> / site : 0.592301

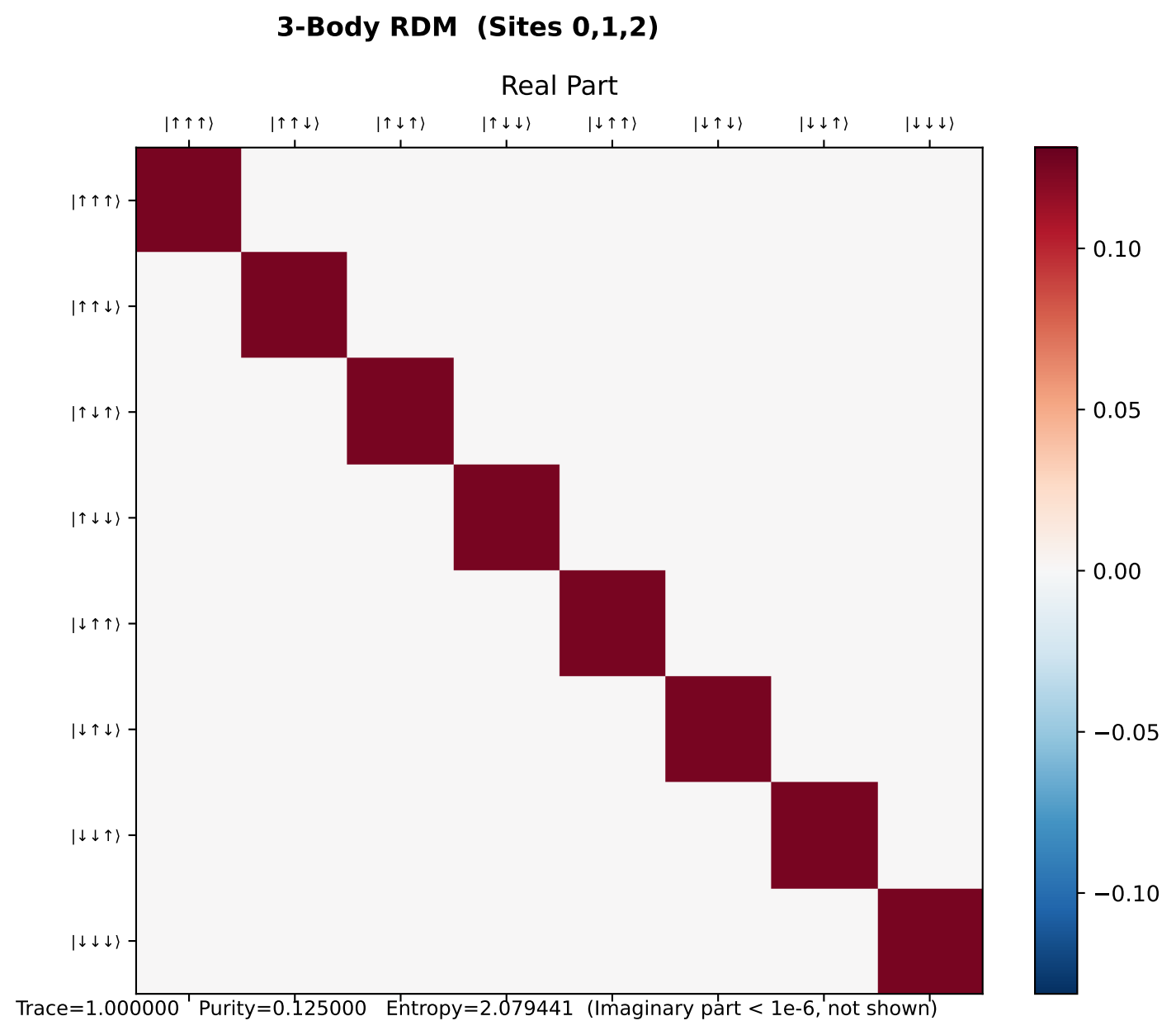
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Body order : 3
Clusters   : 10

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Summary — 3-Body RDMs (Benzene, CAS(6,5))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1,2	0.125000	—	—	—	—	—	STRONGLY MIXED
0,1,3	0.148891	—	—	—	—	—	STRONGLY MIXED
0,1,4	0.125559	—	—	—	—	—	STRONGLY MIXED
0,2,3	0.125559	—	—	—	—	—	STRONGLY MIXED
0,2,4	0.148904	—	—	—	—	—	STRONGLY MIXED
0,3,4	0.125949	—	—	—	—	—	STRONGLY MIXED
1,2,3	0.496207	—	—	—	—	—	MIXED
1,2,4	0.486887	—	—	—	—	—	MIXED
1,3,4	0.499669	—	—	—	—	—	MIXED
2,3,4	0.499782	—	—	—	—	—	MIXED


3-Body RDM (Sites 0,1,2)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,2)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1250000348
Von Neumann entropy	: 2.0794414025
Eigenvalue min	: 1.248858e-01
Eigenvalue max	: 1.250381e-01
Imaginary part	: < 1e-6 (not shown)

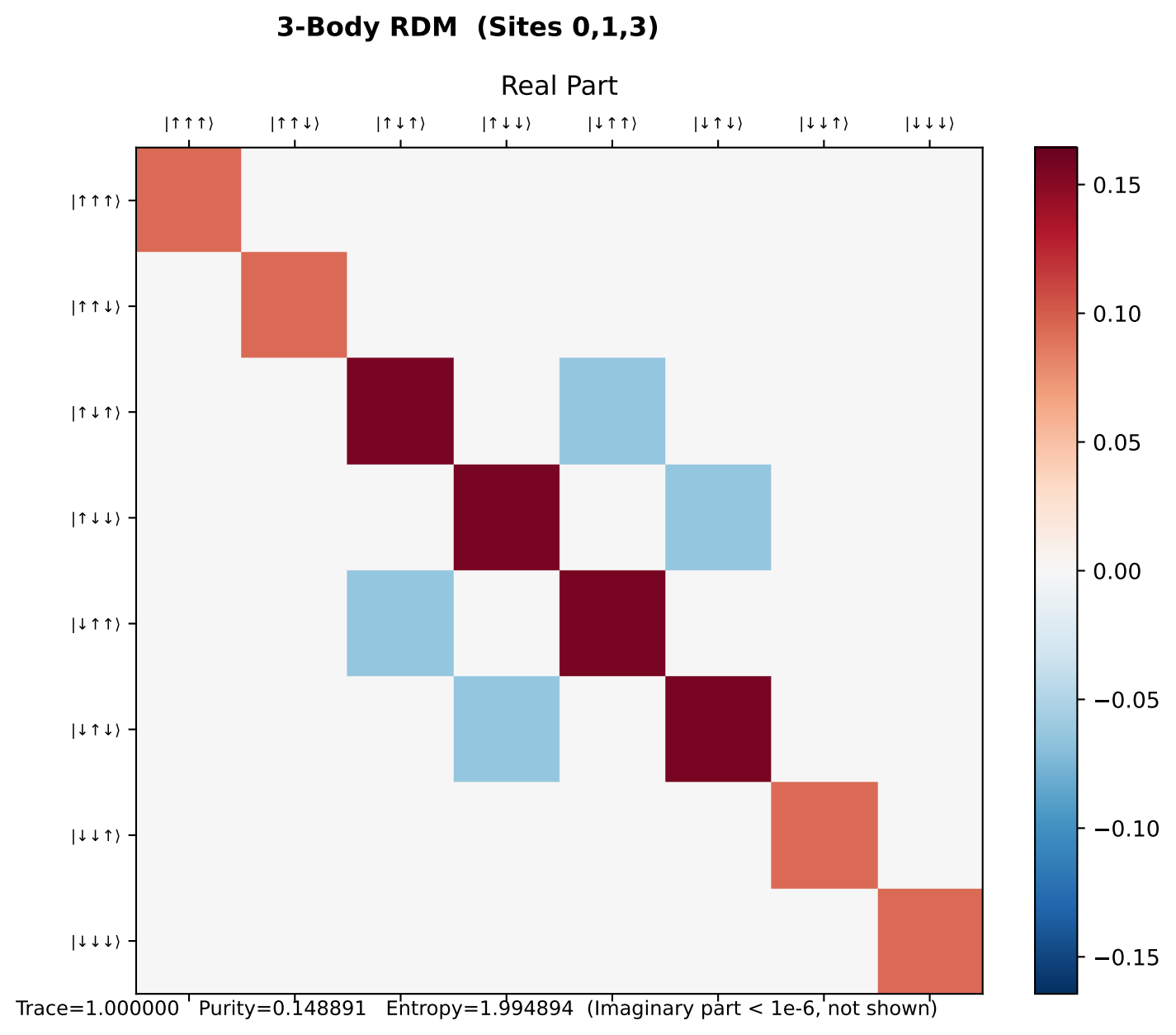
GME Analysis
==========

gamma = Tr(rho^2) = 0.12500003

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


3-Body RDM (Sites 0,1,3)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.148891 Entropy=1.994894 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1488911426
Von Neumann entropy	: 1.9948940631
Eigenvalue min	: 9.344902e-02
Eigenvalue max	: 2.196529e-01
Imaginary part	: < 1e-6 (not shown)

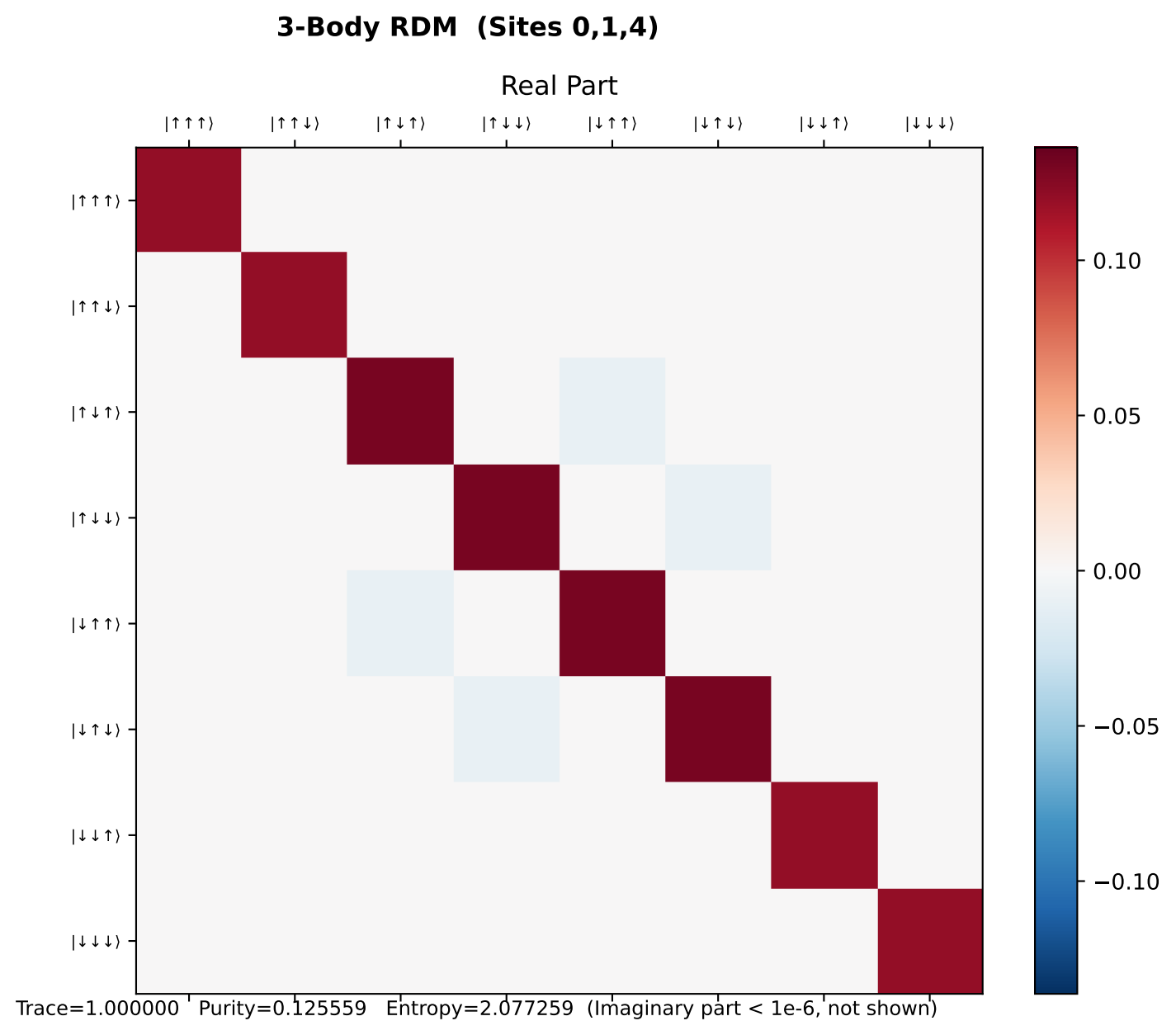
GME Analysis
==========

gamma = Tr(rho^2) = 0.14889114

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


3-Body RDM (Sites 0,1,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125559 Entropy=2.077259 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1255591892
Von Neumann entropy	: 2.0772587151
Eigenvalue min	: 1.201730e-01
Eigenvalue max	: 1.394809e-01
Imaginary part	: < 1e-6 (not shown)

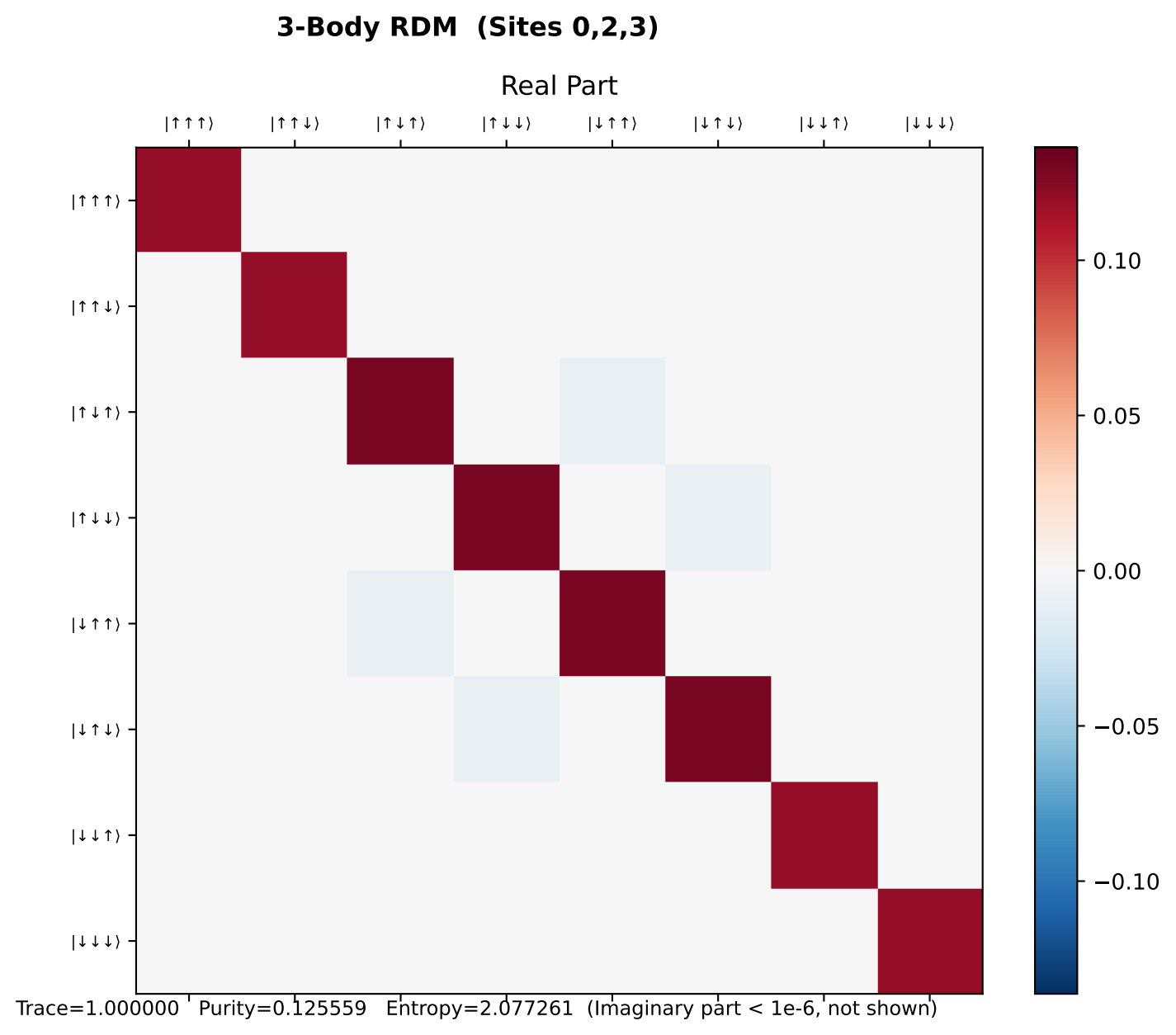
GME Analysis
==========

gamma = Tr(rho^2) = 0.12555919

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


3-Body RDM (Sites 0,2,3)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125559 Entropy=2.077261 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,2,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1255585209
Von Neumann entropy	: 2.0772612935
Eigenvalue min	: 1.201759e-01
Eigenvalue max	: 1.394722e-01
Imaginary part	: < 1e-6 (not shown)

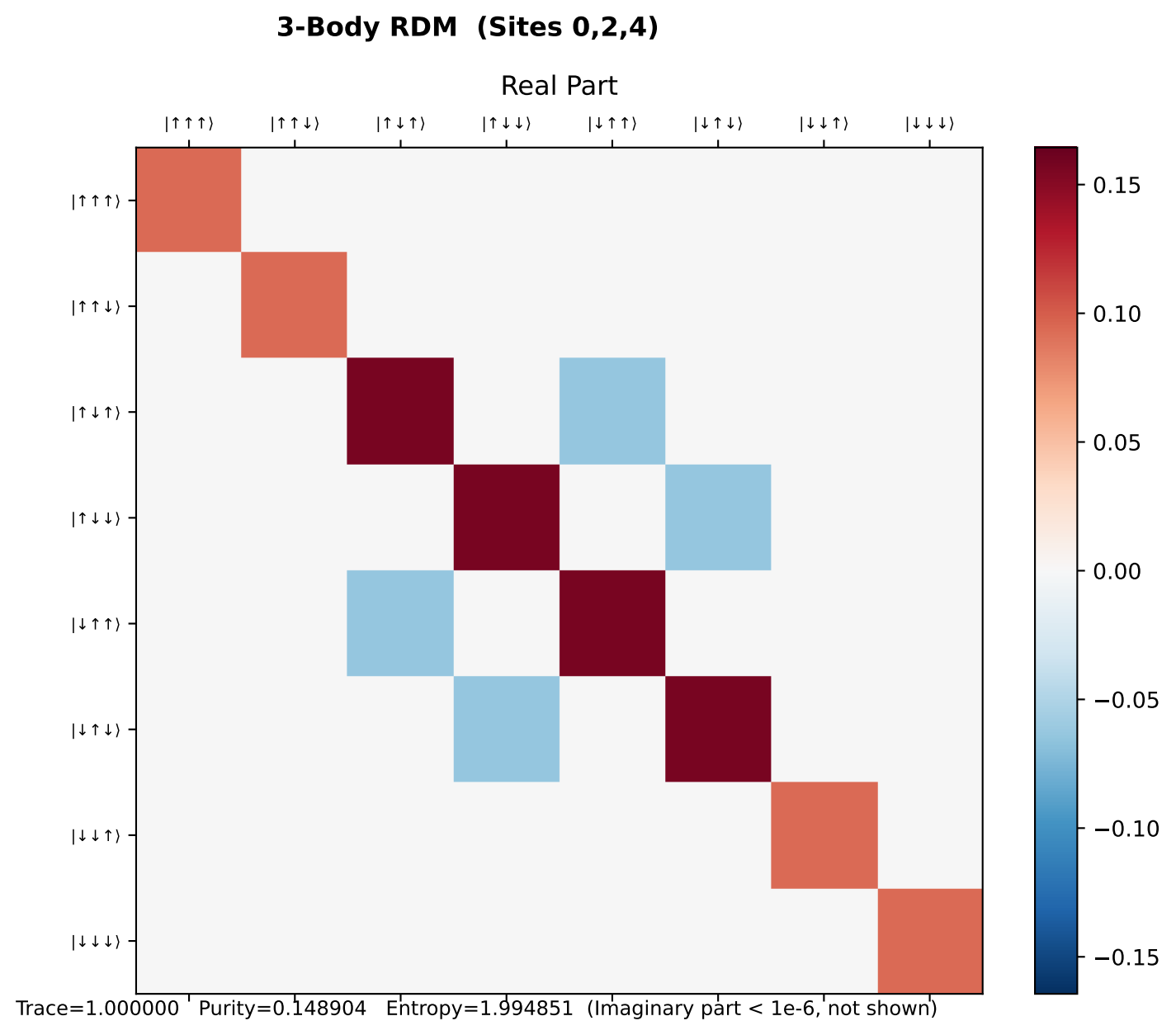
GME Analysis
==========

gamma = Tr(rho^2) = 0.12555852

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


3-Body RDM (Sites 0,2,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.148904 Entropy=1.994851 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,2,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1489040060
Von Neumann entropy	: 1.9948505094
Eigenvalue min	: 9.344053e-02
Eigenvalue max	: 2.196784e-01
Imaginary part	: < 1e-6 (not shown)

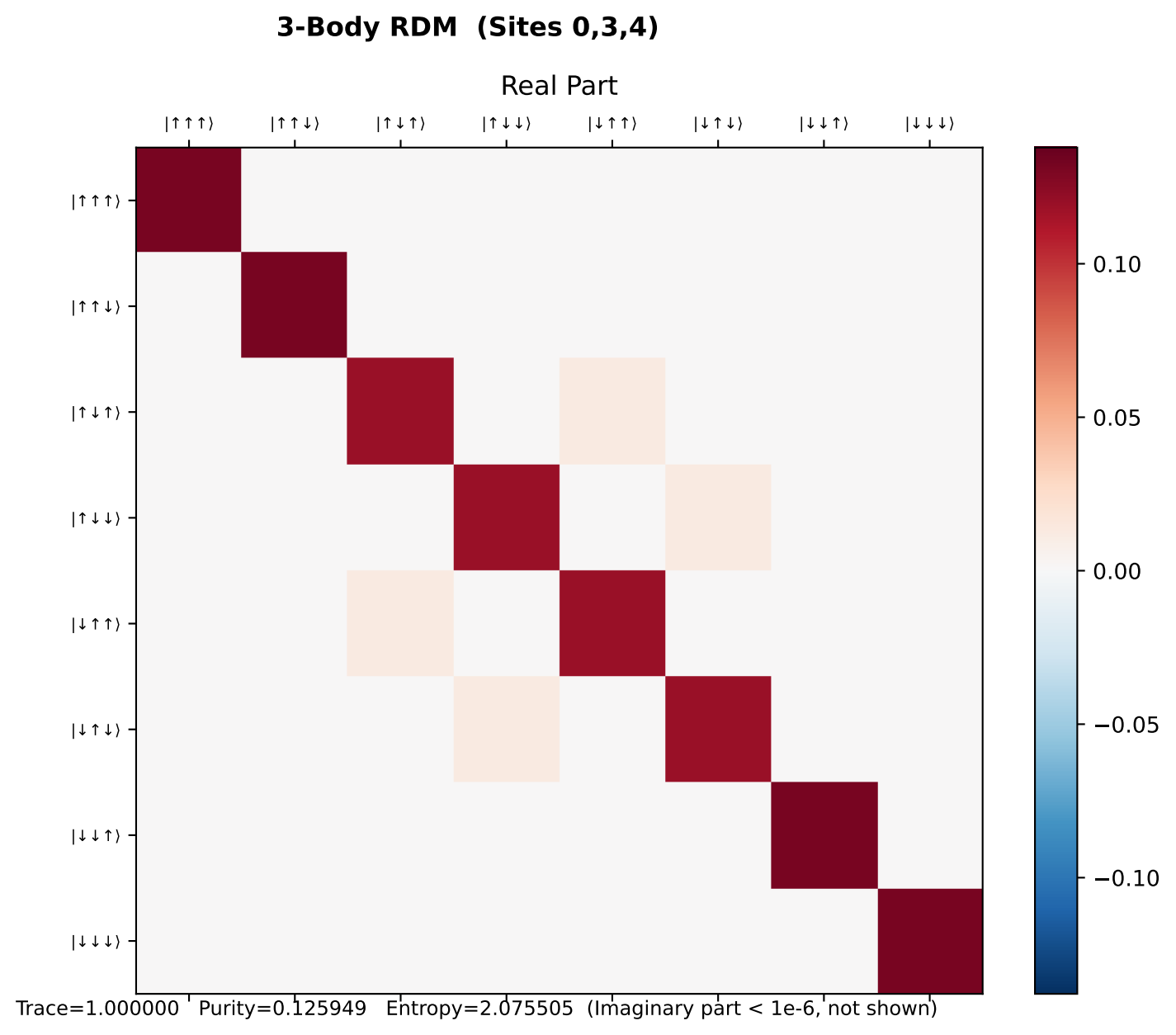
GME Analysis
==========

gamma = Tr(rho^2) = 0.14890401

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


3-Body RDM (Sites 0,3,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125949 Entropy=2.075505 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,3,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1259492576
Von Neumann entropy	: 2.0755048607
Eigenvalue min	: 1.061328e-01
Eigenvalue max	: 1.312891e-01
Imaginary part	: < 1e-6 (not shown)

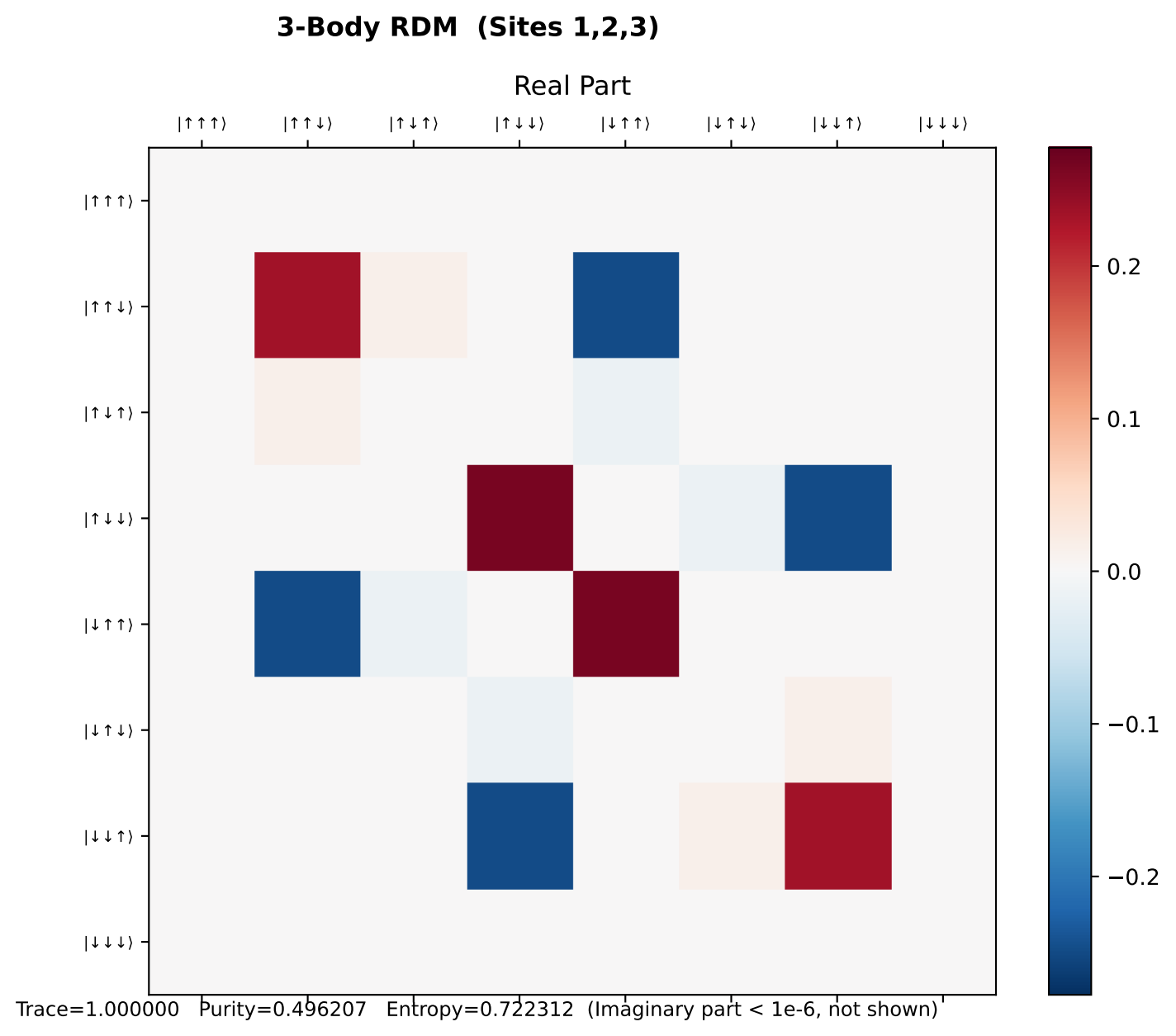
GME Analysis
==========

gamma = Tr(rho^2) = 0.12594926

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


3-Body RDM (Sites 1,2,3)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.496207 Entropy=0.722312 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,2,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.4962065490
Von Neumann entropy	: 0.7223117676
Eigenvalue min	: 6.338489e-04
Eigenvalue max	: 4.980985e-01
Imaginary part	: < 1e-6 (not shown)

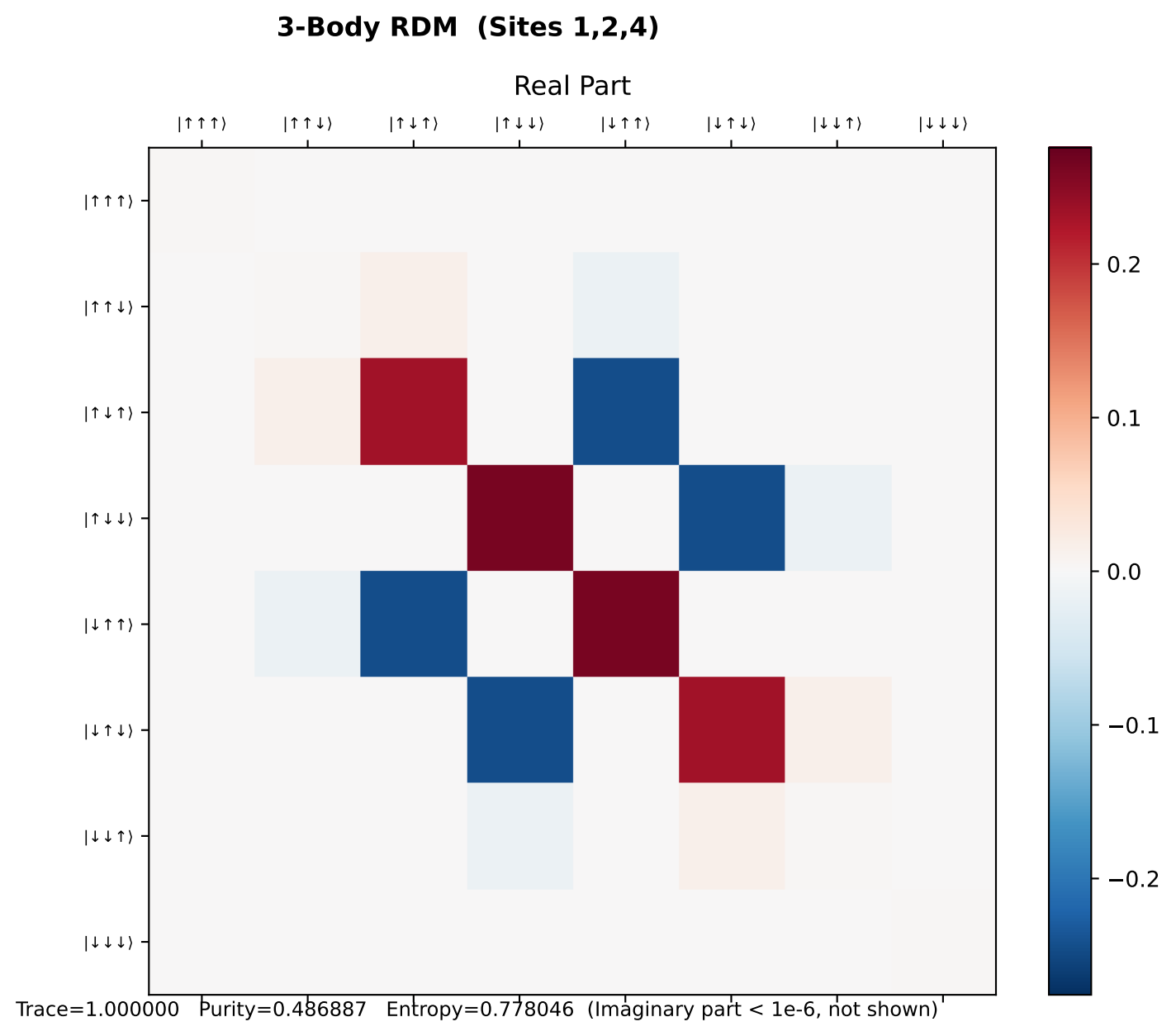
GME Analysis
==========

gamma = Tr(rho^2) = 0.49620655

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


3-Body RDM (Sites 1,2,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.486887 Entropy=0.778046 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,2,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.4868869169
Von Neumann entropy	: 0.7780459582
Eigenvalue min	: 2.204961e-03
Eigenvalue max	: 4.933851e-01
Imaginary part	: < 1e-6 (not shown)

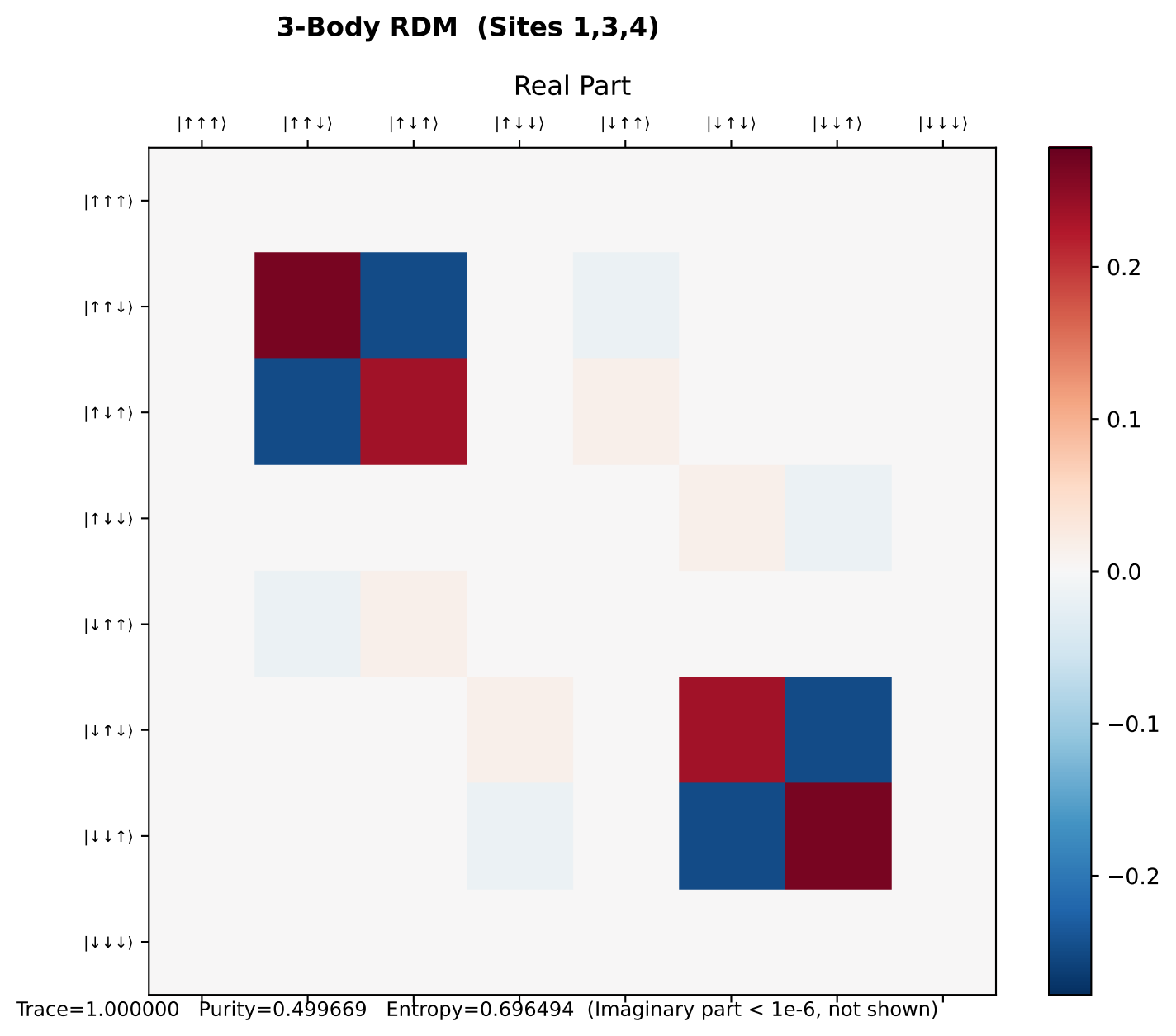
GME Analysis
==========

gamma = Tr(rho^2) = 0.48688692

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


3-Body RDM (Sites 1,3,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.499669 Entropy=0.696494 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,3,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.4996691458
Von Neumann entropy	: 0.6964935362
Eigenvalue min	: 5.515454e-05
Eigenvalue max	: 4.998345e-01
Imaginary part	: < 1e-6 (not shown)

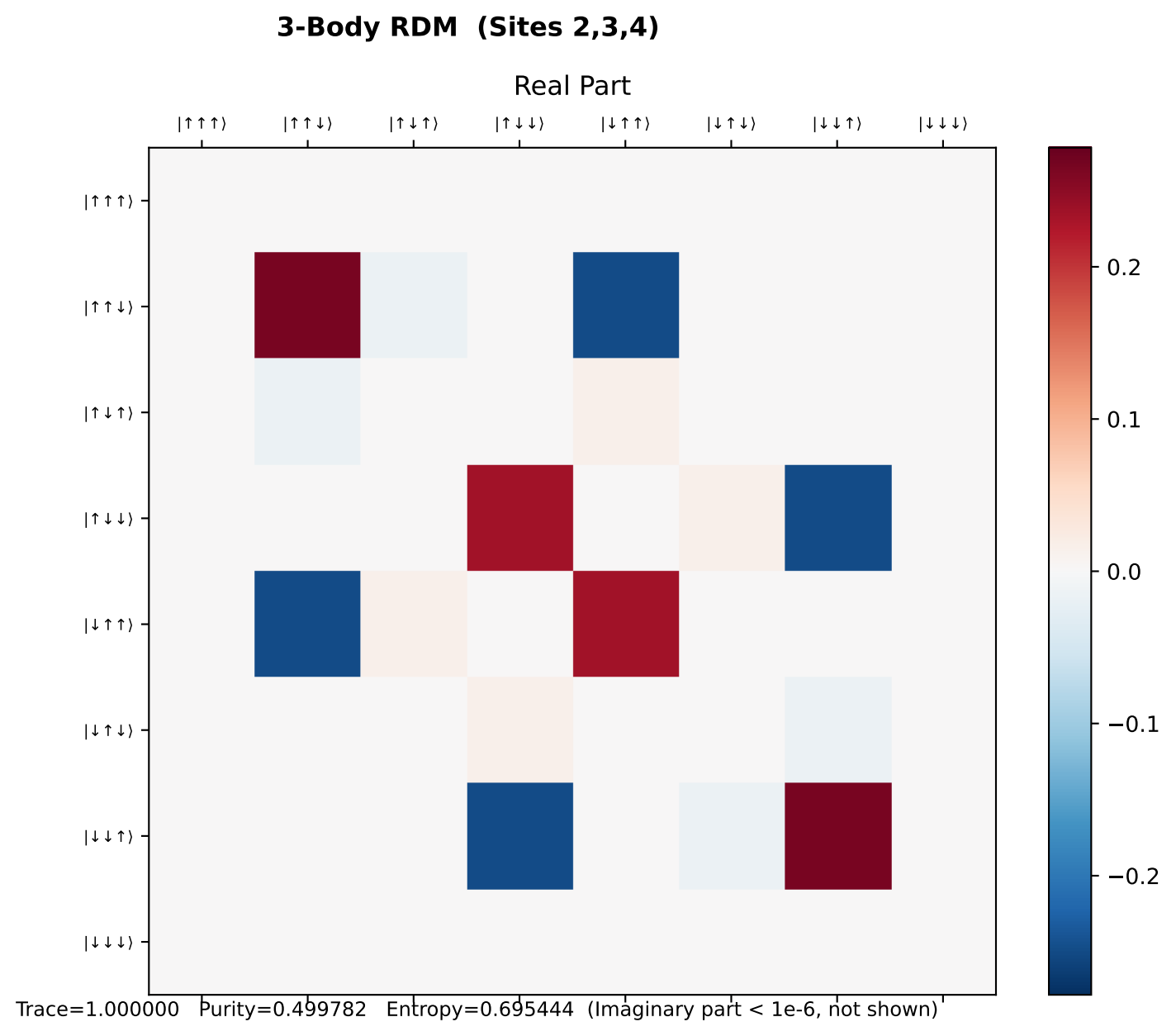
GME Analysis
==========

gamma = Tr(rho^2) = 0.49966915

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


3-Body RDM (Sites 2,3,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.499782 Entropy=0.695444 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 2,3,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.4997819190
Von Neumann entropy	: 0.6954436893
Eigenvalue min	: 3.635211e-05
Eigenvalue max	: 4.998909e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.49978192

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.